

Monetary Policy, Liquidity and Cost of Credit: Evidence from the U.S. Corporate Bond Market

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Abstract

This paper examines the role of bond liquidity in the transmission of monetary policy to the corporate debt market. Using intraday transaction-level data from the Trade Reporting and Compliance Engine (TRACE), we construct daily bond-specific variables, which include yield, credit spread and bid-ask spread. Using high-frequency identified monetary policy shocks from Gurkaynak et al. (2005) and Nakamura and Steinsson (2018), we estimate the effect of a monetary tightening on yields, prices and credit spreads, and we find that the path of future interest rates is quantitatively as important as the federal funds rate in transmitting monetary policy shocks. Furthermore, we document that the yield and credit spread for less liquid bonds—as measured by bid-ask spreads—increases by more as a consequence of a monetary tightening, even when accounting for differences in risk. Lastly, we show that the loading of the liquidity risk factor varies over time and strongly anti-correlates with the slope of the yield curve, suggesting a higher-order nonlinear effect of the liquidity channel.

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1 Introduction

Liquidity in financial markets refers to the ease with which an asset can be traded and converted into cash. In the corporate bond market, liquidity is critical because bonds are traded over-the-counter (OTC), making it more crucial in the pricing of assets compared to equity markets. In fact, transactions in the corporate bond market are not as frequent and take place through specialized financial intermediaries in the form of broker-dealers, which can charge a significant premium on the price. The lower trade frequency, combined with the market power exerted by these dealers, contributes to the structural illiquidity of OTC markets.

This implies that aggregate liquidity conditions shape how securities traded in OTC markets react to changes in monetary policy. In fact, the supply of money and money-like assets plays a pivotal role in amplifying or mitigating liquidity constraints, influencing the pricing of bonds and their response to economic shocks. Given these dynamics, understanding the interaction between liquidity and monetary policy is essential for a comprehensive analysis of the behavior of the credit market.

Monetary shocks can impact the credit spreads of corporate bonds through a variety of channels. For instance, by affecting the risk-free rate, they change the attractiveness of riskless securities relative to corporate debt, which in turn needs to compensate these investors differently. Alternatively, the probability of default can be affected by changes in the policy rate, as the latter has an impact on aggregate demand and therefore firms' profitability and debt sustainability. Investors' risk appetite itself can be impacted by these policy changes.

This study explores the extent to which monetary shocks change the ability to trade corporate bonds in a secondary, decentralized market and how these changes impact the cost of credit. As explored by Ricardo Lagos and Zhang (2020), monetary policy shock change the aggregate amount of cash-like assets in the market, thus affecting trading decisions for investors that operate in illiquid markets.

Our first finding is that monetary policy shocks affect bonds' yield, credit spread and price through both the target rate channel and the term premium channel. Furthermore,

the reaction to monetary policy shocks crucially depends on the liquidity risk of each bond. In particular, the magnitude of the response of each variable is amplified for more illiquid securities. This can be explained by different offloading pressure exerted by a higher interest rate environment on the two risk categories: where flight-to-safety episodes are more likely to happen (i.e. in the High-Yield segment) securities that are traded more frequently are more likely to be re-priced, and hence experience the largest swings in credit spreads.

Additionally, we find that the sensitivity of corporate bond excess returns to liquidity risk varies with the level and slope of the yield curve, suggesting that investors incorporate both current and expected monetary policy stances into bond pricing decisions. This relationship is further reinforced by the observation that liquidity influences how corporate bond spreads react to monetary policy shocks, with less liquid bonds exhibiting a stronger response. These findings underscore the importance of liquidity as a transmission channel for monetary policy and highlight its role in shaping credit market outcomes.

Understanding how monetary policy affects corporate borrowing costs is essential for both financial stability and macroeconomic policy formulation. By examining the interaction between liquidity and monetary policy, this study provides new insights into the mechanisms driving credit market dynamics and the pricing of corporate bonds.

1.1 Literature Review

1.1.1 On Liquidity

Empirical research has developed a range of proxies to measure liquidity in financial markets. Beyond traditional measures such as turnover ratios—the ratio of traded volume to outstanding securities—and bid-ask spreads, the literature has introduced more sophisticated metrics. Notable contributions include price impact measures Kyle (1985), the effective bid-ask spread Roll (1984), and the illiquidity measure of Amihud (2002), which captures the price impact of trading volume by computing the daily average return per dollar traded. Edwards et al. (2007) employ bid-ask spreads to assess direct trading costs, while Bao et al. (2011) introduce a market-adjusted liquidity measure, which accounts

for transitory price movements to reflect varying trading conditions.

Early studies by Amihud and Mendelson (1986) and Amihud (2002) highlight the relationship between liquidity risk, asset prices, and required returns in equity markets. These empirical findings were later formalized in theoretical models such as those of Acharya and Pedersen (2005). In the context of corporate bonds, Chen et al. (2007) and Bao et al. (2011) confirm that liquidity accounts for a substantial share of the variation in corporate bond spreads. The importance of liquidity becomes even more pronounced during periods of financial distress, as illiquidity exacerbates credit spreads, particularly for bonds with higher default risk Lin et al. (2011). Liquidity shocks can also heighten refinancing risk and default probabilities, emphasizing the need for liquidity provisions during crises to prevent market disruptions He and Xiong (2012).

The liquidity of corporate bonds is shaped by multiple factors, including bond characteristics, market structure, and macroeconomic conditions. Empirical evidence suggests that bonds with lower credit ratings and longer maturities tend to be less liquid Houweling et al. (2005). Additionally, callable and structured bonds exhibit lower liquidity due to their complex valuation Helwege et al. (2014). The over-the-counter (OTC) nature of corporate bond trading also contributes to fragmented liquidity, distinguishing it from centralized equity exchanges Bessembinder et al. (2006). Macroeconomic conditions further influence liquidity, with studies showing that liquidity deteriorates and trading volumes decline during financial crises (see Acharya et al. (2013) and Dick-Nielsen et al. (2012)). These patterns were particularly evident during the Global Financial Crisis, when bond markets experienced a sharp contraction in liquidity.

1.1.2 On Monetary Policy and Asset Prices

Monetary policy shocks play a critical role in shaping asset prices and returns by influencing interest rates, risk premia, and expectations about future economic conditions. Classic asset pricing theories suggest that monetary policy primarily affects risk-free rates, which serve as a discounting mechanism for valuing financial assets. However, empirical research has shown that monetary shocks have broader effects on risk premia and asset

valuation, often through unexpected changes in central bank policy decisions. Bernanke and Kuttner (2005) develop an identification strategy to isolate unanticipated monetary policy shocks using expectation revisions, showing that asset prices respond almost immediately to policy changes. Similarly, Gertler and Karadi (2015) provide evidence that contractionary monetary shocks lead to declines in equity prices, driven by increases in the cost of capital and deteriorating expectations about future earnings. Rigobon and Sack (2004) find that monetary policy surprises result in significant declines in stock indices, adopting an identification strategy based on the heteroskedasticity of policy shocks.

Gurkaynak et al. (2005) use high-frequency data around FOMC announcements to isolate and demonstrate that financial markets respond to both current policy rates and central bank communication about the future trajectory of interest rate. This is confirmed by Nakamura and Steinsson (2018) who introduce a method to identify monetary policy news shocks using high-frequency movements in interest rates around FOMC announcements, controlling for information effects.

Monetary policy also affects bond liquidity. Krishnamurthy and Vissing-Jorgensen (2011) argue that higher interest rates can reduce liquidity by increasing holding costs, while quantitative easing (QE) programs improve liquidity by reducing spreads and stimulating trading activity. The risk-taking channel of monetary policy further influences market liquidity and risk appetite Borio and Zhu (2012). Monetary tightening leads to wider credit spreads, reflecting higher default risk and liquidity constraints Hanson and Stein (2014).

2 Data

The main source for trade-level information comes from the Trade Reporting and Compliance Engine (TRACE) database. Each transaction is uniquely identified by its CUSIP code, which links the bond to its issuer and characteristics. The dataset also records the trade date and time in Eastern Standard Time (EST), facilitating an analysis of intraday liquidity fluctuations. In addition to trade timestamps, TRACE specifies the counter-

party type, indicating whether the transaction occurred between two dealers or between a dealer and a customer, and the counterparty side, which denotes whether the trade represented a purchase or a sale for the reporting dealer. The dataset further includes key pricing and volume measures. The trade price represents the transaction price per \$100 face value, while the yield-to-maturity (YTM) is computed based on the bond’s trade price, coupon payments, and time to maturity. The trade volume captures the total par value exchanged in a given trade, with large transactions subject to dissemination caps to limit market impact.

To enrich the dataset with firm-level characteristics, we merge TRACE with the Mergent-FISD database, which provides bond issuance details, including the credit rating at the time of trade and the total amount of outstanding debt for each issuer. Additionally, we integrate financial statement data from COMPUSTAT, focusing on variables related to profitability, liquidity, and leverage. Specifically, we consider operating margin (before depreciation), return on assets, cash ratio, current ratio, debt-to-capital ratio, and debt-to-EBITDA ratio, ensuring a comprehensive view of firm-specific determinants of bond spreads.

We use three different types of high frequency monetary policy shocks. First, we use the *target* and *path* shocks introduced by Gurkaynak et al. (2005) (henceforth, GSS). Target shocks refer to unexpected changes in the federal funds rate itself, capturing surprises in the immediate policy decision. In contrast, path shocks reflect changes in market expectations about the future path of policy, measured by movements in longer-term interest rates that are not explained by the current rate change. On top of these, we also use the monetary policy news shocks introduced by Nakamura and Steinsson (2018), henceforth NS shocks.

2.1 Liquidity

The main empirical section of the paper will use the bid-ask spread Δ^{ba} as a liquidity proxy. Such variable increases with dealers’ market power and lower trading frequency. Hence, higher bid-ask spreads imply stronger dependence on dealers and, therefore, more

illiquid bonds. To compute it, we first, we aggregate the sell prices of sales from dealer to customer for a specific bond i during day t by taking the average. This yields us the daily “bid” price. In formulas:

$$\text{Bid}_{it} = \frac{1}{N_{it}} \sum_{hh \in t} P_{i,hh}(S, D \rightarrow C) \quad (1)$$

where $P_{i,hh}$ represents the price for bond i in transaction hh classified as a dealer-to-customer sale ($S, D \rightarrow C$). The “ask” price, on the other hand, corresponds to the average daily transaction price for customer-to-dealer purchases, given by:

$$\text{Ask}_{it} = \frac{1}{N_{it}} \sum_{hh \in t} P_{i,hh}(B, C \rightarrow D) \quad (2)$$

The bid-ask spread is then computed as the normalized difference between bid and ask prices:

$$\Delta_{it}^{ba} = 2 \times \frac{\text{Bid}_{it} - \text{Ask}_{it}}{\text{Bid}_{it} + \text{Ask}_{it}} \quad (3)$$

2.2 Credit Spreads

To compute credit spreads, we estimate the risk-free yield curve using the model in Nelson and Siegel (1987). For each day t , the risk-free yield at maturity τ is given by:

$$y_t^{rf}(\tau) = \theta_{1,t} + \theta_{2,t} \left(\frac{1 - \exp(-\lambda_t \tau)}{\lambda_t \tau} \right) + \theta_{3,t} \left(\frac{1 - \exp(-\lambda_t \tau)}{\lambda_t \tau} - \exp(-\lambda_t \tau) \right) \quad (4)$$

where the parameters $(\lambda_t, \theta_{1,t}, \theta_{2,t}, \theta_{3,t})$ are estimated using nonlinear least squares from daily yield curve data on Treasuries. The daily credit spread for bond i is then calculated as the difference between the bond’s yield and the estimated risk-free yield at the same maturity:

$$\text{cs}_{it} = \text{YTM}_{it} - y_t^{rf}(\text{TTM}_{it}) \quad (5)$$

where YTM_{it} and TTM_{it} denote the bond’s yield to maturity and time to maturity, respectively.

2.3 Summary Statistics

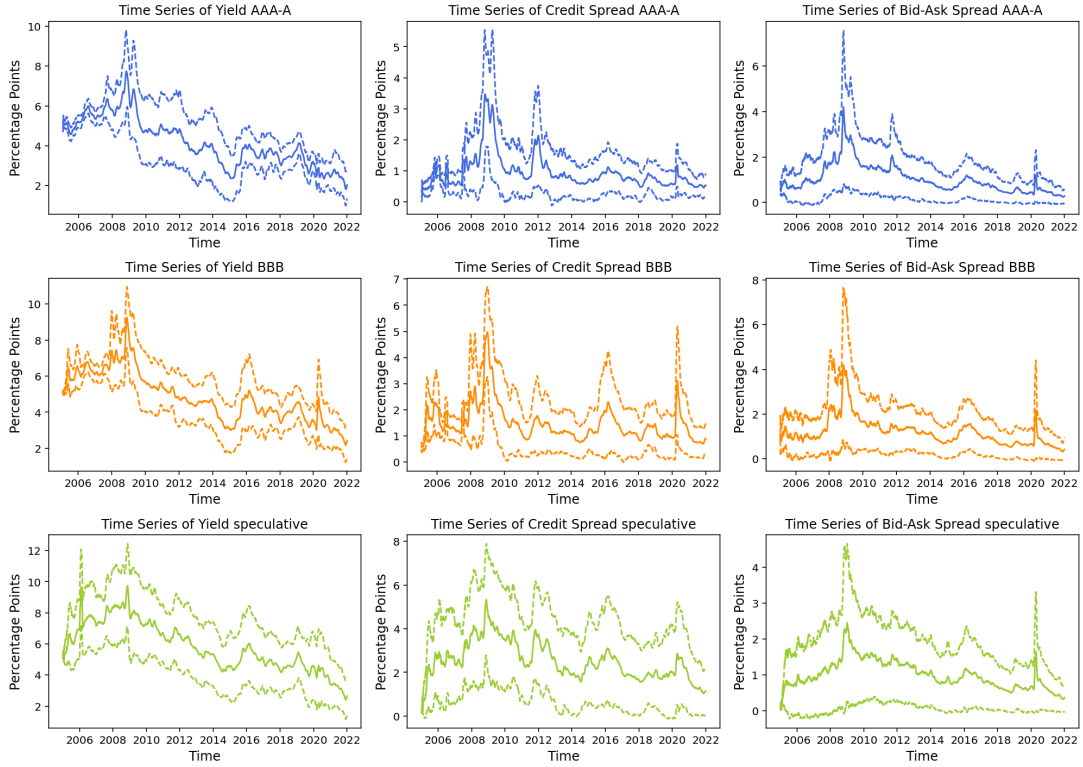


Figure 1: Time series of daily average yield, credit spread and bid-ask spreads across different corporate bond rating categories. Bands represent 1 standard deviation above and below average.

Table 1 reports summary statistics for bond-level yield, credit spreads and bid-ask spread, separately for investment-grade and high-yield bonds. As expected, credit spreads and bid-ask spreads are, on average, higher for high-yield bonds, reflecting their greater credit and liquidity risk. Both measures also exhibit significant right skewness and excess kurtosis, indicating the presence of extreme values. These statistics highlight systematic differences in market characteristics across credit risk classes.

Figure 1, displays the evolution over time of the daily distribution of yields, credit spreads and bid-ask spreads. A clear cyclical pattern emerges, with spikes in credit spreads and bid-ask spreads during periods of financial distress, most notably around the 2008 Global Financial Crisis and the 2020 COVID-19 market shock. Additionally, speculative-grade bonds show a consistently higher and more volatile credit spread compared to investment-grade bonds, reinforcing the notion that riskier bonds face greater borrowing costs and more pronounced liquidity fluctuations

Table 1: Summary statistics by risk class

Variable	Statistic	AAA-A	BBB	Speculative
Yield	Mean	4.090	4.407	5.278
	S.D.	1.730	1.696	2.287
	Q1	3.036	3.286	3.664
	Median	3.968	4.211	4.855
	Q3	5.160	5.372	6.456
	Kurtosis	1.722	2.109	1.004
	Skewness	0.568	0.821	0.947
Credit Spread	Mean	1.013	1.391	2.365
	S.D.	1.039	1.231	2.012
	Q1	0.361	0.543	0.858
	Median	0.730	1.104	1.837
	Q3	1.296	1.855	3.249
	Kurtosis	13.370	7.399	1.728
	Skewness	2.911	2.143	1.365
Bid-Ask Spread	Mean	1.060	1.072	0.931
	S.D.	1.178	1.093	0.967
	Q1	0.263	0.308	0.258
	Median	0.645	0.725	0.597
	Q3	1.433	1.492	1.294
	Kurtosis	7.325	7.859	7.209
	Skewness	2.272	2.229	2.150

3 Response to Monetary Policy Shocks

3.1 Event Study Analysis

We begin our analysis assessing the effects of monetary policy changes on a set of corporate bond market outcome variables. In particular, we want to estimate what is the impact of an increase in the policy rate or in the future path of policy rates on yields, credit spreads and liquidity. It is important to stress the difference between the two different channels-the current policy rate and the expected path of it-as corporate bonds are known to have high duration compared to other assets, like, for example, stocks. Hence the future expected trajectory of interest rates, and their influence on long term rates, is of particular interest when pricing corporate bonds.

The event study consists of estimating the following regression:

$$y_{it} = \alpha + \beta_1 \text{ffr}_t + \beta_2 \text{slope}_t + \alpha_i + \alpha_r + \epsilon_{it} \quad (6)$$

where t is restricted to be in the set of FOMC announcement days, $y \in \{\text{Yield, Credit Spread, } \Delta^{ba}\}$, and $\alpha_i, \alpha_t, \alpha_r$ are bond, time and risk fixed effects ($r \in \{\text{HY, IG}\}$). The independent variables of interest are the policy rate, i.e. the federal funds rate ffr_t , and the slope of the yield curve, computed as the 10 year-3 months spread on Treasuries. This last variable is meant to capture the future expected path of interest rates. Due to endogeneity between the dependent and independent variables, we use the full set of high-frequency shocks (i.e. GSS target, GSS path and NS) as instruments for the policy variables in Equation 6. Table 2 shows the result of this estimation. Interestingly, the magnitude of the impact of shocks to the slope of the yield curve are consistently higher effects on every outcome variable, suggesting that long term rates, and therefore the future expected path of interest rates, has a higher loading in pricing corporate bonds.

The next step in our analysis is to assess whether monetary policy has differential effects on the outcome variables depending on the liquidity of the bonds. To this end, we

Table 2: Estimation results for Equation 6. Standard errors in parentheses.

	Yield	Cs	Ba C	Price	Volume
ffr	1.396*** (0.069)	0.521*** (0.061)	0.420*** (0.039)	-7.165*** (0.411)	-59825.216* (30736.495)
slope	1.646*** (0.052)	0.866*** (0.047)	0.650*** (0.028)	-1.271*** (0.330)	140833.444*** (21603.080)

need to estimate the effect of the interaction between policy variables and liquidity as in the following equation:

$$y_{it} = \alpha + \beta_1 \Delta_{it}^{ba} + \beta_2 \text{ffr}_t + \beta_3 \text{slope}_t + \beta_4 \text{ffr}_t \times \Delta_{it}^{ba} + \beta_5 \text{slope}_t \times \Delta_{it}^{ba} + \alpha_i + \alpha_t + \alpha_r + \epsilon_{it} \quad (7)$$

where $y \in \{\text{Yield, Credit Spread}\}$. A major econometric challenge to the specification of Equation 7 is that liquidity changes on announcement day as a consequence of the monetary policy shock, as revealed in Table 2. Furthermore, there are again endogeneity concerns due to the fact that bonds yields and credit spreads might influence bonds' tradeability. These observations invalidate the use of contemporaneous bid-ask spreads as a proxy for bonds' intrinsic liquidity. To overcome these issues, we introduce $\overline{\Delta}^{ba}$, computed as the average bid-ask spread from 30 days prior to the FOMC announcement up to the day before:

$$\overline{\Delta}^{ba}_{it} = \frac{1}{30} \sum_{s=1}^{31} \Delta_{it-s}^{ba} \quad (8)$$

The liquidity proxy constructed in this way is therefore orthogonal to the monetary policy shock, and they classify bonds in the cross section based on their intrinsic tradeability over a reasonably long horizon.

Table 3 shows the results of the IV estimation of Equation 7. It is clear that less liquid bonds, i.e. those with higher $\overline{\Delta}^{ba}$, are more affected by monetary policy shocks.

Table 3: Estimation results for Equation 7. Standard errors in parentheses.

	Yield	Credit Spread	Price
$\text{ffr} \times \overline{\Delta^{ba}}$	0.198*** (0.030)	0.230*** (0.029)	-2.935*** (0.211)
$\text{slope} \times \overline{\Delta^{ba}}$	0.210*** (0.036)	0.093*** (0.034)	1.898*** (0.195)

3.2 Dynamic Responses and the Liquidity Channel

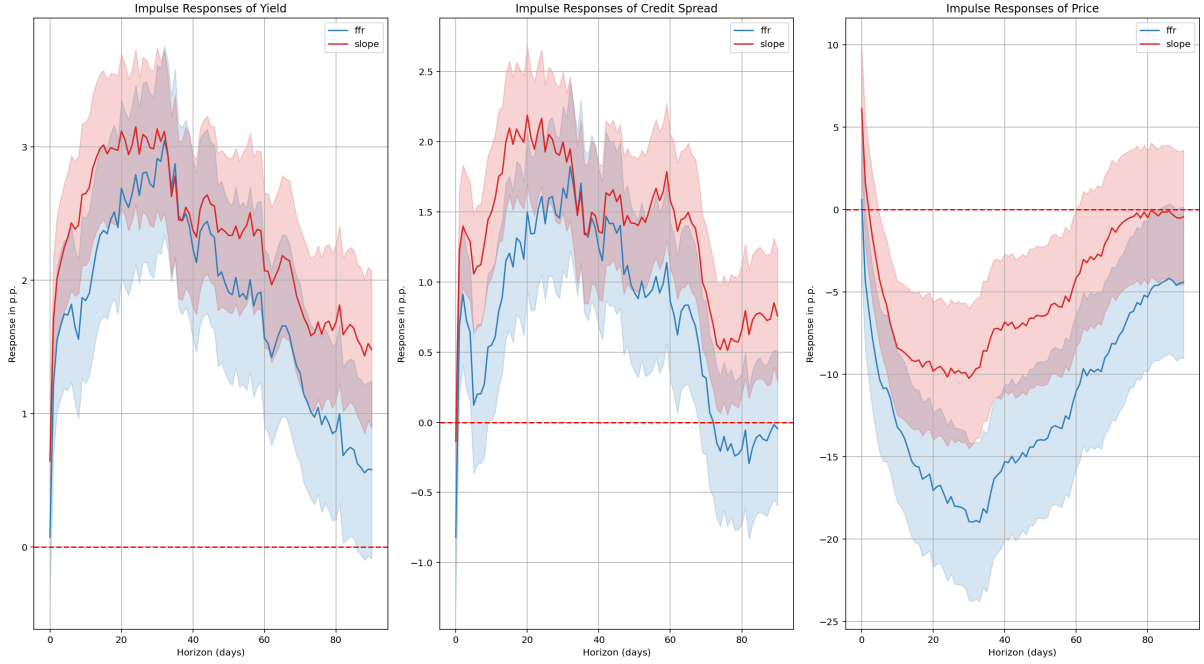


Figure 2: Coefficients $\beta_{1,h}$ and $\beta_{2,h}$ over an horizon of 30 trading days.

The dynamic effects of monetary policy are estimated through a version of Equation 6 across different horizons:

$$y_{it+h} = \alpha_{i,h} + \beta_{1,h}\text{ffr}_t + \beta_{2,h}\text{slope}_t + \gamma_h \mathbf{x}_{t-1} + \epsilon_{it+h} \quad (9)$$

The results are displayed in Figure 2, which reports the set of coefficients $(\beta_{1,h}, \beta_{2,h})_{h=1}^{30}$. These confirm that shocks to the long term rates, manifested through the slope coefficient, seem to be larger and more persistent than increases in the policy rate.

Next, we are interested in assessing whether bonds with different degrees of liquidity are affected differently by monetary policy shocks. To this end, we focus on Investment

Grade (IG) bonds, in order to remove the confounding effects of names with high default risk, and we split these into 2 portfolios, sorted by liquidity (Low, High). This categorization is performed first by computing a liquidity proxy that is exogenous to the monetary shock, that is the average of these two measures on the 30 days before the FOMC announcement. Bonds whose liquidity proxy is above (below) the cross sectional median of turnover (bid-ask spread), it is labeled as "High", and "Low" otherwise. We then estimate the following regression:

$$y_{i,t+h} = \alpha_i(h) + \sum_p \beta_p(h) \cdot [\mathbf{1}(i \in p) \times \varepsilon_t^{mp}] + \gamma'(h)\mathbf{x}_{it} + \epsilon_{it} \quad (10)$$

where ε_t^{mp} represents the high-frequency identified monetary policy shock and $\mathbf{x}$ are aggregate controls.

According to the local projection methodology introduced by Jordà (2005), the impulse response function of a shock at horizon h is the coefficient $\beta_p(h)$. In fact, the impulse response function of variable y at horizon h to a monetary policy shock of size v is formally defined as:

$$IRF(h, t, v) = \mathbb{E}_t[y_{t+h} \mid \varepsilon_t = v] - \mathbb{E}_t[y_{t+h} \mid \varepsilon_t = 0] \quad (11)$$

where the operator $\mathbb{E}_t$ is the expectation conditional on information available at time t . Conditional on the econometric model specified in Equation 10, Equation 11 boils down $\beta_p(h) \cdot v$.¹ Hence, this framework allows us to quantify how monetary policy differentially affects credit spreads based on liquidity and risk exposure, providing insights into the transmission mechanisms of monetary policy in corporate bond markets.

Figure ?? and 3 shows the IRF generated by Equation 10. A shock to the current policy rate target (i.e., a "GSS target" shock) leads to an increase in the credit spread,

¹This argument can be generalized to nonlinear settings. Assume that y depends on the shock and other aggregates according to the (smooth) functional form $y_{t+h} = \Phi(\mathbf{x}_t, \varepsilon_t)$, then the Taylor expansion around $\mathbf{0}$ is:

$$y_{t+h} = \Phi(\mathbf{0}, 0) + \nabla_{\mathbf{x}}\Phi \cdot \mathbf{x}_t + \frac{\partial\Phi}{\partial\varepsilon}\varepsilon_t$$

where $\nabla_{\mathbf{x}}$ represent the entries of the gradient of Φ corresponding to the vector $\mathbf{x}$. Hence, Equation 11 becomes $IRF(h, t, v) = \frac{\partial\Phi}{\partial\varepsilon}$, which can be estimated with a linear model as in Equation 10.

yields and a drop in prices that is concentrated on the less liquid segment of the market, with the low-liquidity-IG bonds responding the most. In fact, their yields increase on average by almost 20 basis points, their credit spreads by 15 basis points and their prices drop by 50 basis points. Low-liquidity-HY bonds also significantly react, although not as much as their IG counterparts. In contrast, highly liquid bonds are considerably less sensitive to the shock across all liquidity categories. This suggests that, in the face of immediate monetary policy tightening, investors prioritize liquidating and illiquid securities, accepting greater price concessions in the process.

Liquidity does not seem to play a major role in the responses to shocks to the expected path of future short-term interest rates (i.e., "GSS path" shocks), where the main differences across names appears to be driven by risk alone. In fact, IG bonds react on impact to these shocks, experiencing an increase by 10 b.p. in yields, 5 b.p. in credit spreads and a decrease in price by 30 b.p. HY bonds, on the other hand, are slower to respond, showing a delayed dynamics of the same magnitude as their safer counterparts. A similar configuration is observed in response to monetary policy news shocks (NS), although in this case liquidity plays a significant role. In fact, less liquid bonds display a larger movement in response to the shock, exactly as in the case to the shock to the target rate.

To quantify the liquidity channel of monetary policy on credit spreads, we proceed in two steps. First, we apply the same local projection methodology introduced in the previous paragraph to estimate the partial effect of the monetary policy shock on liquidity. Hence, we first estimate:

$$\Delta_{i,t+h}^{ba} = \alpha_i(h) + \sum_{r \in \{HY, IG\}} \beta_r(h) \cdot [\mathbf{1}(i \in r) \times \varepsilon_t^{mp}] + \gamma(h)' \mathbf{x}_t + \epsilon_{it} \quad (12)$$

Figure 4 presents the impulse response functions estimated from Equation 12. The effect on Bid-Ask spreads are in line with what it expected. After an increase in the target rate, Bid-Ask spreads increase by about 5 basis points for High Yield bonds and by 15 basis points for Investment Grade bonds, and both stay permanently positive in the

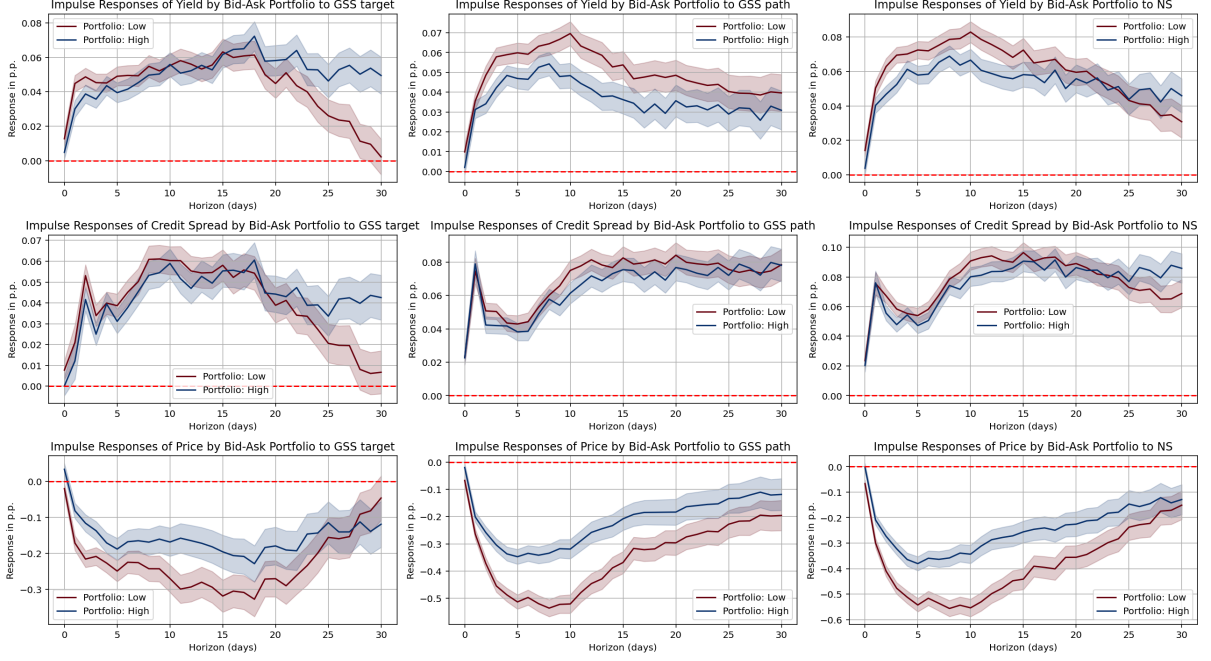


Figure 3: Impulse responses of bond yield, credit spread and price to monetary policy shocks. Bonds are divided in portfolios sorted by bid-ask spread.

considered horizon. A similar pattern emerges when considering news shocks, whereas interest rates path shocks have a much milder effect and, although persistent. These responses suggest that monetary policy shocks make the corporate bonds market less liquid, and particularly so for Investment Grade securities.

4 Higher-order liquidity channel of monetary policy

In this Section we document that monetary policy has higher order effects on the corporate bonds market. In particular, the monetary policy shocks change the liquidity risk factor loading on the credit spread paid by corporate security. Firstly, we define the liquidity risk factor, $lrf^\Delta = \log \Delta^{ba}$.

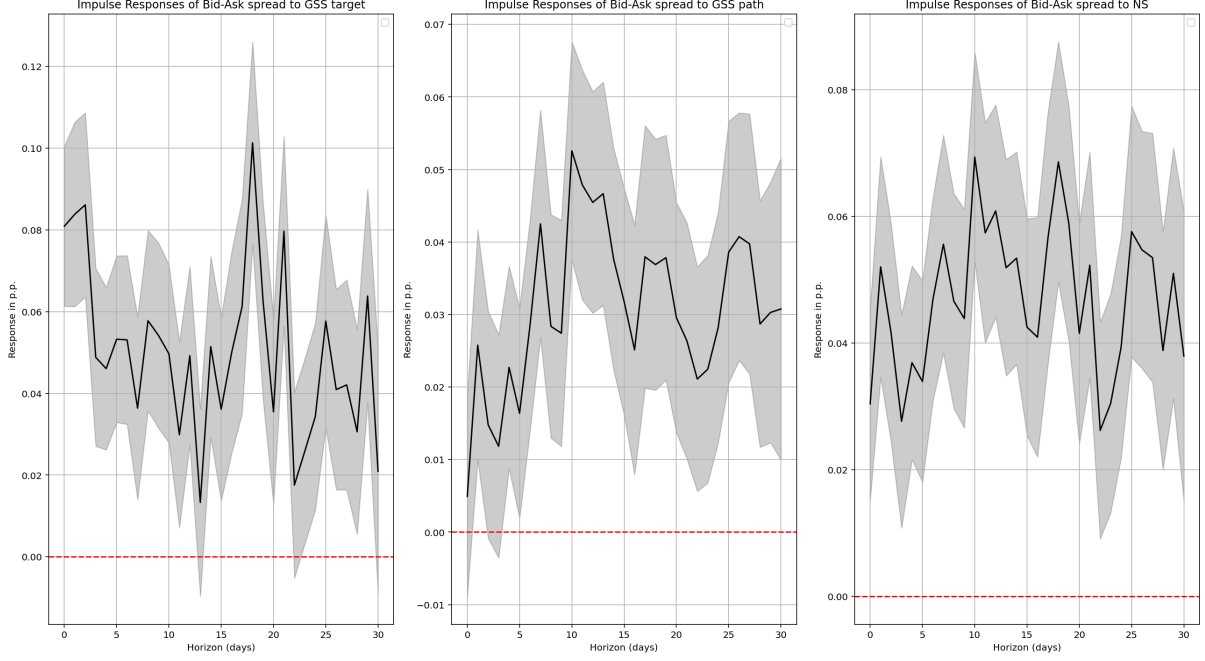


Figure 4: Impulse responses of credit spreads to monetary policy shocks, bonds are divided in low and high liquidity according to their turnover.

4.1 Liquidity Risk factor Loading

To assess the time varying nature of the relationship between liquidity risk and credit spreads, for each month t_m we estimate the following panel regression model:

$$cs_{i,t} = \alpha_{t_m}^R + \beta_{t_m}^{liq} lrf_{it} + \gamma'_{t_m,0} \mathbf{x}_t^{agg} + \gamma'_{t_m,1} \mathbf{x}_{i,t}^{fr} + \epsilon_{it} \quad t \in t_m \quad (13)$$

where lrf_{it} represents a liquidity risk factor, and $\mathbf{x}_t^{agg}$ and $\mathbf{x}_{i,t}^{fr}$ denote aggregate and firm-specific control variables, respectively. Aggregate controls include the level and slope of the yield curve, the VIX index, and the ICE-BofA spread between high-yield and investment-grade bonds. Firm-specific controls capture profitability, liquidity leverage.

Given potential endogeneity concerns, we employ an instrumental variable (IV) approach, using the average number of trades in the previous 30 days as an instrument for lrf_{it} . This variable is strongly correlated with liquidity but is exogenous to *current* credit spreads.

4.2 Results

Figure 5 plots this coefficient against the slope of the yield curve, measured as the spread between the 10 year Treasury yield and the 3 month Treasury rate. The chart hints at a strong anti-correlation between the two variables. To quantify this relationship, we run the regression:

$$\beta_{t_m}^{\text{liq}} = \alpha + \phi_r r_{t_m}^{1 \text{ mo}} + \phi_{\text{slope}} \text{slope}_{t_m} + \phi_{\text{stock}} \text{Stock}_{t_m} + \phi_{VIX} VIX_{t_m} + \nu_{t_m} \quad (14)$$

The results are displayed in Table 4.

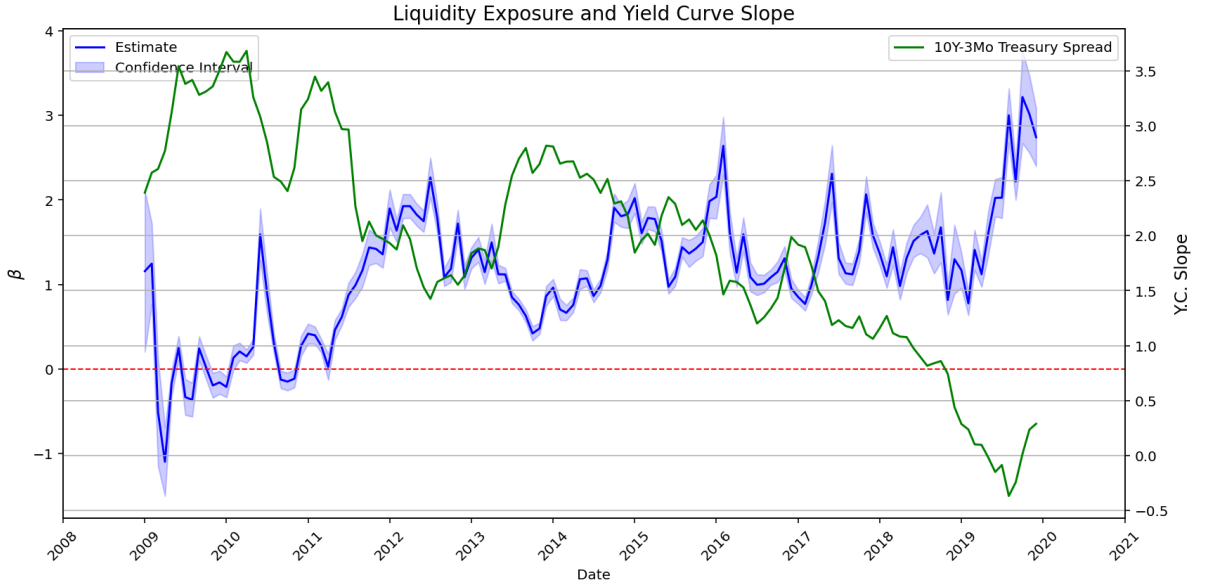


Figure 5: The loading on the liquidity risk factor plotted against the 10 year - 3 month spread highlights a negative comovement between the two variables. Only pre-Covid sample is displayed.

5 Conclusion

This research highlights the crucial role of liquidity in determining how monetary policy shocks affect corporate credit spreads. The empirical results provide evidence that more liquid bonds display lower credit spreads all else equal, but they also experience smaller responses to monetary policy shocks. This suggests that liquidity dampens away the effects of rate hikes. These findings have implications for policymakers and mar-

	<i>Dependent variable: β^{liq}</i>
const	2.916*** (0.210)
r^{1Mo}	-0.561*** (0.102)
slope	-0.784*** (0.074)
SP500	0.000* (0.000)
VIX	-0.007 (0.006)
Observations	120
R^2	0.581
Adjusted R^2	0.567
Residual Standard Deviation. Error	0.449
F Statistic	39.912***

Table 4: Results of OLS estimation of Equation 14. Note: *p<0.1; **p<0.05; ***p<0.01

ket participants, emphasizing the need for liquidity considerations in financial stability assessments and monetary policy design.

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